

Research papers

Temporal inductive biases in hourly flood forecasting: a comparative analysis of recurrent, attention-based, and state-space neural networks

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ARTICLE INFO

This manuscript was handled by Dan Lu, Editor-in-Chief, with the assistance of Phong Le, Associate Editor

Keywords:

Deep learning
Hourly forecasting
Runoff
Flood
Mamba
Benchmark

ABSTRACT

Daily-resolution flood forecasts often encounter difficulties in capturing rapid runoff fluctuations over short time periods. To address this limitation, moving to hourly-scale modeling is essential, yet identifying which deep learning architectures possess the optimal temporal inductive biases for such high-frequency dynamics remains an open challenge. In this study, we conduct a comprehensive comparative analysis of recurrent, attention-based, and state-space neural networks, benchmarked against simple layered structures on discharge-assisted forecasting tasks across 516 CAMELS catchments. The results indicate that the Patch Transformer achieves the best predictive performance. This suggests that patch-based processing acts as a superior inductive bias by smoothing high-frequency noise and extracting local temporal semantics, which effectively captures long-term routing and hydrological dependencies better than point-wise attention. While the traditional LSTM maintains robust performance, the Mamba model demonstrates a promising balance between accuracy and computational efficiency. Furthermore, through an interpretability analysis combining attention weights with Integrated Gradients (IG), we reveal that attention models successfully capture physical hydrological patterns, such as time-lagged responses and the dominance of dynamic meteorological drivers over static attributes. This work offers theoretical insights into how architectural choices determine short-term forecasting capability under diverse hydro-climatic regimes.

1. Introduction

Floods rank among the most frequent and destructive natural disasters worldwide, with the incidence of flood-related events having more than doubled over the past two decades (Hirabayashi et al., 2013; Tellman et al., 2021; N. Wang et al., 2026). Effective runoff forecasting models, which predict the timing and magnitude of floods, are crucial for mitigating flood risks, safeguarding lives and property, and supporting scientific water resource management and ecological conservation (Chen et al., 2025; Jongman et al., 2015). Fundamentally, runoff and flood forecasting encompass spatiotemporal modeling challenges

(Troin et al., 2021; Zhang et al., 2023), characterized by substantial variations in temporal scales and methodological frameworks.

From a temporal perspective, forecasts are broadly categorized into medium- to long-term and short-term to real-time types. Medium- to long-term forecasts, with lead times ranging from weeks to months (Hong et al., 2016), are vital for managing sediment transport, analyzing and mitigating droughts, informing water resource allocation and discharge decisions, and preserving biological habitats and reproductive cycles. In contrast, short-term and real-time forecasts feature lead times from several to dozens of hours (H. Wang et al., 2025) and are critical for ensuring reliable hydropower operation, issuing flash flood warnings,

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<https://doi.org/10.1016/j.jhydrol.2026.135727>

Received 12 December 2025; Received in revised form 21 April 2026; Accepted 19 May 2026

Available online 28 May 2026

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and activating disaster reduction systems. Several operational flood forecasting systems, such as the global GloFAS from the Copernicus Emergency Management Service, utilize models like the European Medium-Range Weather Forecast to assess flood risks for the upcoming days (Alfieri et al., 2013). Similarly, Google's Flood Hub project offers water level forecasts for up to seven days in over 80 countries (Nearing et al., 2024). However, these systems predominantly operate at a daily resolution, which presents significant limitations in capturing real-time streamflow dynamics. The coarse temporal granularity of daily inputs is particularly problematic given that intense flash floods can complete the rainfall-runoff process within hours (Assouline, 2013), thereby posing severe threats to downstream areas. Aggregating and averaging monitoring data to a daily scale often obscures critical short-term fluctuations. Consequently, daily-scale data offer limited utility for next-day predictions (Gauch et al., 2021). To date, hourly-scale runoff and flood forecasting has been primarily explored through localized case studies, such as in South Korea's Han River (Nguyen et al., 2022), the Appalachian region of the United States (Smith et al., 2011), and Tunxi, China (Malik et al., 2024). A comprehensive large-scale evaluation at an hourly resolution remains lacking, underscoring an urgent need for more refined forecasting capabilities.

From a methodological perspective, forecasts can be divided into conceptual models, dynamic models, and data-driven models (Troin et al., 2021). Conceptual models simulate runoff by establishing mathematical or statistical relationships grounded in watershed hydrological cycle theory (Miao et al., 2022). However, their dependence on empirical knowledge and susceptibility to parameter uncertainty often restrict their application in large-sample hydrological studies. Dynamic models, which employ numerical methods to simulate the rainfall-runoff process, can represent the dynamic evolution of floods but are computationally intensive (Q. Xu et al., 2024). This high computational demand often hinders their ability to deliver timely warnings for large watersheds. In contrast, data-driven models learn patterns and relationships directly from historical observational data (Reichstein et al., 2019). Among these, deep learning models—particularly recurrent neural networks (RNNs) with recursive structures like LSTM and GRU—have gained considerable traction in flood forecasting. This includes LSTM-based sequence-to-sequence (S2S) models and encoder-decoder architectures (Feng et al., 2020; Xiang et al., 2020). The appeal of LSTM lies in its cell state mechanism, which naturally mirrors the storage-release dynamics inherent in hydrological processes, effectively capturing state evolution over time or Markovian properties (An and Ouyang, 2025; Khatun et al., 2024). Despite their black-box nature and potential for overfitting, LSTMs have proven highly reliable for global flood forecasting, owing to their rapid response and efficiency in processing large datasets (Zhang et al., 2024). A key limitation of LSTM-based models, however, is their inability to explicitly model direct, weighted connections between arbitrary time steps in a sequence. This shortcoming can be addressed by Transformer models (Yin et al., 2022). The Transformer architecture, a major recent innovation in deep learning, relies solely on self-attention mechanisms to capture global dependencies within sequences (Vaswani et al., 2023). Its parallel processing nature offers greater flexibility and scalability for handling large-scale data. Transformers have demonstrated state-of-the-art performance across diverse domains, including language translation (Wang et al., 2019), image recognition (Dosovitskiy et al., 2021), text generation (Bang et al., 2023), and protein design (Chandra et al., 2023). Nevertheless, their efficacy in hydrological tasks remains debated. Some studies (Xu et al., 2023) have highlighted the potential of Transformers for global flood forecasting, while others (J. Liu et al., 2024; Rasiya Koya and Roy, 2024) have identified inherent limitations in the standard Transformer module for hydrological simulation, arguing that any perceived advantages over LSTM may be marginal and more attributable to scaling effects than architectural superiority.

This gap suggests that the potential of Transformer-based architectures for large-scale hydrological modeling remains underexplored,

primarily due to the absence of a rigorous and fair comparative framework. Furthermore, since flood forecasting is inherently a sequence prediction problem (Gauch and Lin, 2020), it is more appropriate to evaluate Transformer variants that are specifically engineered for time series analysis, rather than the original architecture designed for natural language processing. Several such advanced Transformer models—including Informer (Zhou et al., 2021), Autoformer (Wu et al., 2022), and Patch Transformer (Nie et al., 2023), have been validated in cases of traffic, finance, and power forecasting. In this study, we adapt these specialized models for hydrology, perform a systematic comparative analysis, and investigate the behavior of the attention mechanism and its hydrological interpretations from both a temporal and variable perspective.

Furthermore, recent advances have introduced the Mamba model into hydrology (Y. Wang et al., 2025; Xu et al., 2026), which is built upon a structured state space model (SSM) core. Empirical evidence shows that Mamba achieves a favorable balance between performance and computational efficiency compared to Transformers in domains such as computer vision and natural language processing (Gu and Dao, 2024). This efficiency stems from the SSM's use of convolutional computation to capture sequential dependencies while obviating the need for explicit hidden states, thereby enabling parallel training and yielding nearly linear scaling in computational complexity with sequence length. To assess its applicability in hydrology, we implemented and evaluated a hydrological adaptation of Mamba, namely S-Mamba (Wang et al., 2024), within our study.

We present a comprehensive comparative study of deep learning models for hourly-scale runoff and flood forecasting. Building upon existing daily-scale literature, this research systematically evaluates refined forecasting capabilities across a large sample of catchments. To provide a rigorous evaluation, we explicitly distinguish between two operational settings: discharge-assisted forecasting, which incorporates historical streamflow observations as inputs to capture high-frequency routing dynamics, and pure rainfall-runoff forecasting, which relies strictly on meteorological forcings and static attributes. While our primary experiments and interpretability analyses focus on discharge-assisted forecasting, we also provide a comparative baseline under the pure rainfall-runoff setting. This comparative study encompasses models representing diverse architectural paradigms: simple linear models with layered components, widely adopted recurrent models, increasingly prevalent self-attention models, and the newly emerging selective state-space models.

The main contributions of this work are summarized below:

- (1) A standardized comparative framework was established with strict experimental controls to fairly evaluate how different architectural inductive biases perform across diverse hydroclimatic regimes at an hourly scale.
- (2) The Mamba was evaluated for high-frequency forecasting, highlighting its potential to balance predictive accuracy and operational computational efficiency compared to mainstream paradigms.
- (3) A multi-perspective interpretability analysis was conducted by cross-validating attention weights with IG. Despite local discrepancies, their trends consistently reveal how these models capture physical hydrological processes.

2. Methods

2.1. The framework of runoff and flood forecasting based on deep learning benchmark

While deep learning has achieved considerable success in daily-scale runoff and flood forecasting, many operational scenarios demand predictions at a finer temporal resolution. This study presents a large-scale comparative analysis of deep learning models for hourly-scale runoff

and flood forecasting. The overall framework, depicted in Fig. 1, establishes a fine-grained benchmark based on deep learning and comprises two main stages:

Stage I: Multi-Model Forecasting and Comparative Evaluation. Historical meteorological forcing data, streamflow records, and static catchment attributes serve as inputs to establish predictive relationships between various models and the target watershed's streamflow. Multiple evaluation metrics are then used to comprehensively assess and compare the streamflow prediction performance of each model. (See Fig. 2.).

Stage II: Interpretability Analysis of Self-Attention Mechanisms. This stage focuses on a detailed examination of models with self-attention architectures (e.g., Transformers) to explore the links between attention scores and hydrological processes. We perform a quantitative analysis of how attention scores vary across historical time windows, using Patch Transformer as a representative case, and how they are distributed across different input variables, using Informer as an example.

2.2. Benchmark models

For the rainfall-runoff task, our research summarizes recent deep learning models in order of technological iterations, categorizing them into four main types and selecting the most representative models to establish a benchmark set.

2.2.1. Dlinear (layered structures)

The study of flood prediction modeling using deep learning frameworks can be traced back to feed-forward neural networks (Hsu et al., 1995). This type of network consists of an input layer, hidden layers, and an output layer. It utilizes learnable weight parameters to directly simulate various nonlinear relationships between inputs and outputs. Dlinear is a simple neural network model with a layered stacking structure (Zeng et al., 2022) and reports indicate that it demonstrates performance comparable to various complex models across different prediction tasks. This model uses a moving average kernel to decompose the raw input data into trend and seasonal components. Each component is then used as input to a single linear layer, and the two outputs are summed to obtain the final prediction result. Dlinear features no

redundant model design and, according to Occam's Razor (Blumer et al., 1987), can be used to validate the effectiveness of other models.

2.2.2. LSTM (recurrent structures)

Compared to feed-forward neural networks, LSTM is a type of recurrent neural network structure with memory units (Sherstinsky, 2020; Z. Wang et al., 2026) and belongs to a sequential structure, making it particularly useful for simulating natural systems with sequential characteristics, such as watersheds. LSTM avoids the issues of gradient explosion or vanishing and has achieved significant success in the hydrological field, demonstrating capabilities comparable to state-of-the-art numerical forecasting systems in daily-scale global flood forecasting. The model structure can be found in reports from literature such as (Kratzert et al., 2018).

We explicitly chose the standard hourly LSTM over the Multi-Timescale LSTM (MTS-LSTM; Gauch et al., 2021) to ensure an "apples-to-apples" structural comparison. Our objective is to rigorously isolate architectural inductive biases, which strictly requires identical look-back windows (96 h) across all models. Evaluating 96-hour models against an MTS-LSTM designed for > 365-day extended memory would confound architectural benefits with input-length advantages. Furthermore, Gauch et al. (2021) demonstrated that standard LSTMs perform comparably to MTS-LSTM in strict hourly tasks, confirming its suitability as a robust baseline.

2.2.3. Transformers (self-attention structures)

The uniqueness of the Transformer model lies in its self-attention module. The formula for calculating attention scores is provided in Eq. (1). This module transforms each input token into three different types of encoded information: 'queries' that inquire about the relevance of other inputs, 'keys' that respond to the relevance of other inputs, and 'values' that contain the actual meaning of the input. Attention scores are computed for each input based on this information.

It is notable that self-attention mechanisms process inputs in parallel, which gives transformer models an advantage when handling long time series. This allows them to consider the relationships between sequence information from any non-adjacent positions. In some cases, two non-adjacent positions may have a strong relationship, whereas models with serial structures like LSTM rely on the output of the previous step for the current input. This not only faces the issue of forgetting long sequence information but also suffers from cumulative regression errors and lower inference speed.

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

However, as a module designed for natural language processing, the original transformer models exhibit some redundant designs in hydrological applications, such as the masking mechanism for word vectors and the exponential complexity of attention, leaving significant room for improvement. Therefore, in addition to Rainfall runoff transformer (RRformer, RRformer is based on basic transformer architecture), other four specifically improved transformer models for time series prediction (RRformer, Patch TST, Informer, Autoformer) are selected to test of the potential of transformer models in fine-grained watershed flood forecasting.

RRformer is the first model to conduct rainfall-runoff modeling based on Transformer architecture on a large sample (Yin et al., 2022). It maintains the original Transformer structure while applying linear transformations to the input, supplemented with learnable position embeddings that are fed into both the encoder and decoder (Garza et al., 2024).

PatchTST, on the other hand, utilizes instance normalization operators to segment each channel's univariate data and perform patch operations. These patches are used as input tokens, employing channel-independent shared embeddings and weights for effective

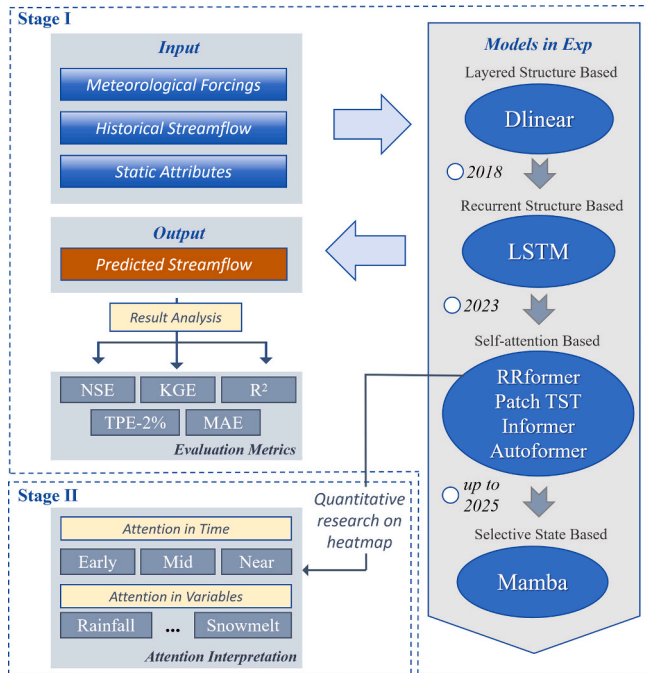


Fig. 1. Deep learning-based fine-grained catchments runoff and flood forecasting benchmark set and analysis framework.

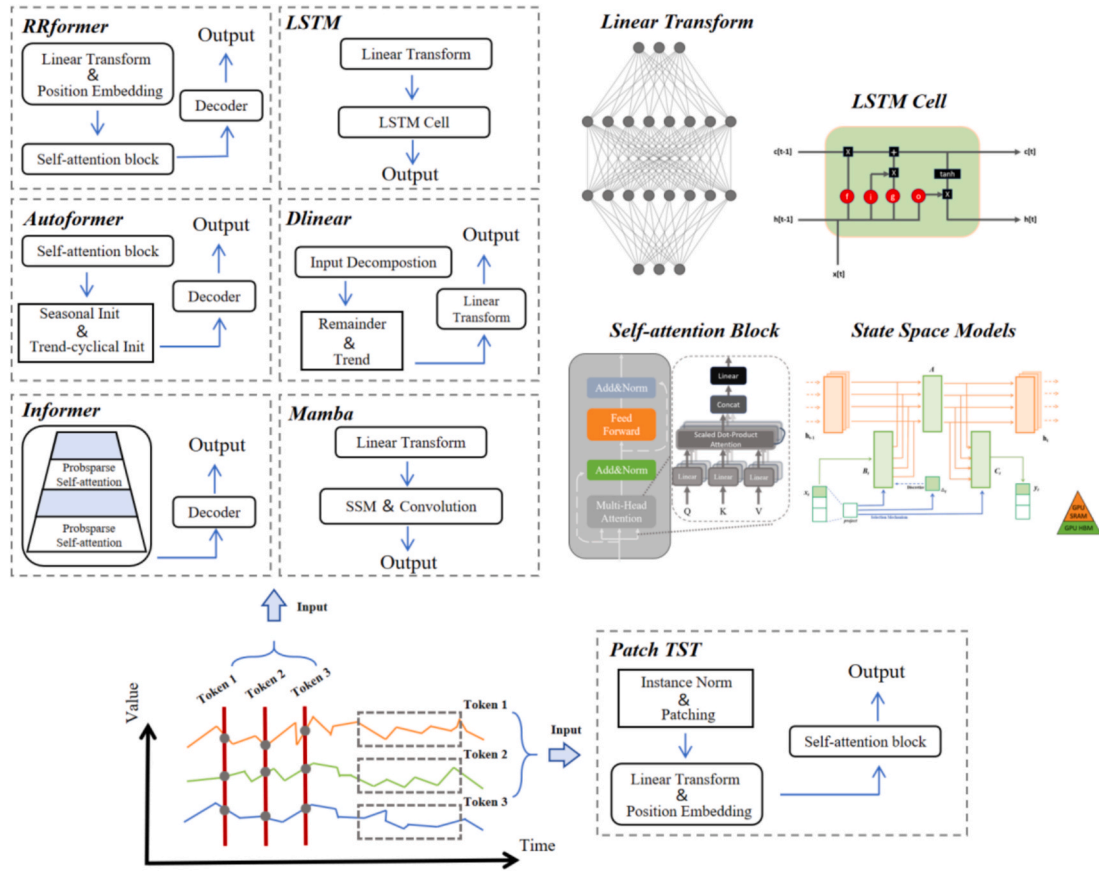


Fig. 2. Simplified models overview. Except for Patch TST, the input tokens for other models have the same shape.

representation learning.

Informer introduces a sparse attention mechanism with distillation technology to effectively extract the most important feature portions and reduce computational complexity.

Autoformer decomposes sequential features into seasonal and trend components, replacing the point-wise self-attention mechanism in the original Transformer with an auto-correlation mechanism, enabling series-wise long-term predictions.

2.2.4. Mamba (selective structures)

Although Transformer models can effectively capture patterns and dependencies within sequences, their parallel computation has a time complexity of $O(n^2)$ where n is the sequence length. This leads to significantly increased computational costs as the sequence length grows, hindering feasibility in long-term sequence predictions. To address this issue, Ahamed (Ahamed and Cheng, 2024) proposed a selective state space architecture called Mamba for time series prediction tasks, achieving accuracy comparable to that of Transformers across multiple tasks.

The Mamba model is characterized by its Structured State Space Model (SSM), which represents system behavior through a set of hidden variables called 'states,' allowing it to effectively capture dependencies in temporal data. The SSM uses a set of differential equations (Eq. (2)) to describe internal system changes, where $x(t) \in \mathbb{R}^D$ can map the sequence through the latent state $h(t) \in \mathbb{R}^N$ to $y(t) \in \mathbb{R}^N$, with $A \in \mathbb{R}^{N \times N}$, $B, C \in \mathbb{R}^{N \times D}$ being learned matrices. Another set of equations (Eq. (3)) describes the relationship between the latent states and the output sequence.

Unlike recurrent neural network models, the SSM employs a method known as Zero-Order Hold (ZOH) to enforce discretization, turning continuous parameters into discrete ones. The discrete SSM is highly

flexible, with computational overhead varying linearly or near-linearly with sequence length. Thanks to its linear properties, it supports both recurrent calculations and convolutional computations, and it incorporates hardware-aware parallel algorithms into its recurrent mode, unlike RNNs, where each computation relies solely on the previous hidden state and the current input.

$$\begin{aligned} h(t)' &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) \end{aligned} \quad (2)$$

In this study, an improved Mamba model called S-Mamba (Wang et al., 2024) is used for rainfall-runoff modeling tests, extracting the correlations and temporal dependencies of variables and inputting them into the basic Mamba module and feedforward network for runoff prediction.

After the SSM discretization, the continuous input sequence is discretized into time steps of Δ . And $\bar{A} = \exp(\Delta A)$, $\bar{B} = (\Delta A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B$. Due to the transition from the continuous form (Δ, A, B, C) to the discrete form $(\bar{A}, \bar{B}, C)$, it can be effectively computed using linear recursive methods.

$$\begin{aligned} h_t &= \bar{A}h_{t-1} + \bar{B}x_t \\ y_t &= Ch_t \end{aligned} \quad (3)$$

3. Dataset and setting

3.1. Dataset

The experimental data were derived from the publicly available CAMELS dataset (Addor et al., 2017), which provides catchment attributes and meteorological observations across a large sample of watersheds. The original dataset comprises 671 catchments across the United

States, with drainage areas ranging from 4 km² to 25,000 km², and includes meteorological time series, catchment characteristics, and streamflow observations for each basin.

To ensure data consistency, we excluded catchments with substantial discrepancies in area estimates resulting from different calculation methods, retaining only those with drainage areas under 2,000 km². This resulted in a final set of 516 catchments, aligning with the benchmark established by Kratzert et al. (2019).

We further extended the modeling framework to an hourly resolution. All timeseries for the 516 catchments were aligned to Coordinated Universal Time (UTC). Streamflow observations were obtained through the USGS Instantaneous Values REST API (U.S. Geological Survey, 2021), which serves hourly discharge measurements within the USGS Water Information System. Meteorological forcing data were sourced from the NLDAS-2 product (Xia et al., 2012), providing hourly records from 1979 onward. These variables were spatially averaged over each catchment. In addition, we incorporated 27 static attributes for each watershed, covering soil properties, climate indices, vegetation characteristics, and topographic metrics. These attributes were derived from maps, remote sensing products, and climatological datasets, all of which are broadly available across the contiguous United States and other regions, though with varying accuracy. A full listing of variables is provided in the [supplementary materials](#).

All experiments were performed on a Linux system equipped with four NVIDIA V100 GPUs, each with 32 GB of memory. The meteorological forcings and static attributes used in this study are summarized in [Table S1](#) of the [supplementary documents](#).

3.2. Model setting and experimental setup

To establish a rigorous comparative framework, the datasets for all 516 catchments were chronologically divided into training, validation, and testing periods. Specifically, the training set comprises 21,000 hourly samples (starting from October 1, 1989), the validation set consists of 3,000 samples (starting from February 23, 1996), and the test set spans 6,000 samples (starting from June 27, 2001). For catchments lacking historical records, the starting timestamp was rolled forward chronologically to the first available data point while strictly maintaining the sequential length requirements.

To ensure data quality, missing values and quality-control gaps in the USGS instantaneous records were imputed using linear interpolation. While the validation set spans a relatively short temporal window (~4 months), we utilized this 7:1:2 split ratio to maximize the length of the continuous testing set (~8 months), ensuring robust evaluation across a broader range of seasonal hydro-climatic events. Furthermore, as demonstrated in the [Appendix](#) (see [supplementary Fig. S4](#)), there is no significant data distribution shift among the training, validation, and testing sets, confirming the representativeness of this partition strategy.

The input sequence length (look-back window) was strictly set to 96 h across all models, while the prediction horizon was set to the subsequent 24 h. For models utilizing an encoder-decoder architecture (Informer, RRformer, Autoformer), the decoder input sequence length was set to 48 h.

To integrate the watershed characteristics, the 27 static attributes were replicated across the temporal dimension and directly concatenated with the dynamic meteorological forcings at each time step to form a unified input vector. Ablation experiments regarding the inclusion or exclusion of these static attributes were conducted; the detailed results are provided in the [Appendix](#) ([Table S3](#)).

3.3. Training procedure and hyperparameter optimization (HPO)

To ensure a fair comparison and avoid claims confounded by unequal tuning budgets (Hosseini et al., 2024), we conducted a systematic budget-matched HPO. As summarized in [Table 1](#), we defined a shared search space and executed 100 uniform random search trials for each

Table 1

Summary of the hyperparameter search space for HPO and the selected configurations in the manuscript.

Category	Hyper-parameter	Search space (range)	Selected value
Common	Learning rate	1e-4, 5e-4, 1e-3	1e-4
	Input sequences length	48, 72, 96, 120, 168	96
	Hidden dimension	64, 128, 256, 512	512
	Dropout rate	0, 0.05, 0.1, 0.2	0.05
	Batch size	32, 64, 128, 256	256
	Loss function	RMSE, NSE, KGE	RMSE
LSTM	Initial forget gate bias	-3, -1, 0, 1, 3	1
	Layers	1, 2	1
Transformer Family	Number of heads	4, 8, 16	8
	Encoder layers	1, 2, 3	2
	Decoder layers	1, 2, 3	1
	Decoder input length	24, 48, 72	48
Patch_TST	Patch length	4, 8, 16, 24	8
Dlinear	Moving average window	10, 25, 40	25
Autoformer	Moving average window	10, 25, 40	25
Mamba	State dimension	8, 16, 32	16

model architecture. The parameter configuration yielding the minimum average validation loss was selected. A subsequent analysis of the HPO trials distribution (detailed in the [Appendix](#)) confirmed that our selected hyperparameters performed well above the median across the 100 random searches, indicating robust parameter suitability for the respective architectures.

During the training phase, all models were optimized using the Adam optimizer with the “gelu” activation function (Hendrycks and Gimpel, 2023). The maximum number of training epochs was set to 100. To prevent overfitting, a strict early stopping strategy was implemented: training was halted if the validation loss failed to improve for 5 consecutive epochs. Specific architectural settings were configured as follows: Informer utilized distillation operations; Autoformer and Dlinear’s sequence trend decomposition sliding window used a step size of 25; Patch TST employed fully connected network in the head layer to integrate the information of different channels; Mamba utilized forward-backward bidirectional encoding layers; and Reversible Instance Normalization (RevIN) was applied where appropriate to mitigate potential data distribution shifts (Fang et al., 2022). Other default hyperparameters were kept consistent as detailed in [Table 1](#).

4. Results and discussion

4.1. Location distribution of the best NSE models in different catchments

All models in the benchmark generated continuous predictions for the next 24 h (detailed formulations of the evaluation metrics are provided in the [supplementary materials](#)). [Fig. 3](#) presents the Nash-Sutcliffe efficiency (NSE) values across different lead times, with subplots (a) to (f) statistically summarizing the best-performing model for each catchment at 1-, 3-, 6-, 9-, 12-, and 24-hour horizons. Each subplot includes a pie chart in the lower left corner, illustrating the proportion of catchments for which a given model achieved the highest NSE, with the best models explicitly labeled.

For the shortest lead time (1-hour), the Dlinear model demonstrated a dominant advantage, achieving the best performance in 76.7 % of the catchments, which suggests that a simple linear model is adequate for ultra-short-term forecasting. However, its superiority diminished rapidly as the forecast horizon increased. The PatchTST model emerged as the most effective for longer lead times of 3, 6, 9, 12, and 24 h, attaining the highest NSE in 91.3 %, 74.3 %, 67.3 %, 64.7 %, and 53.4 % of the catchments, respectively. The LSTM model was the second-best

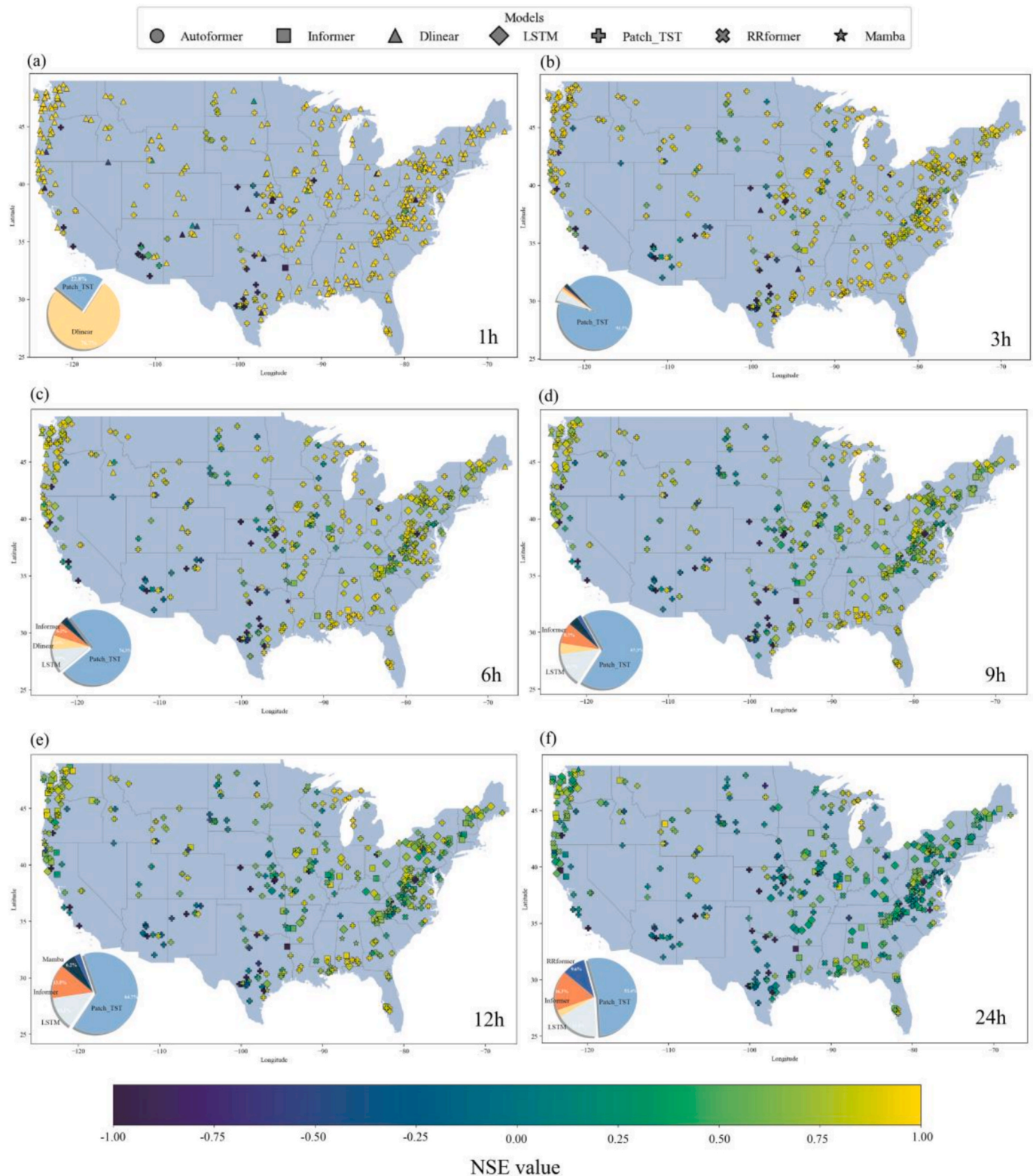


Fig. 3. The optimal NSE values of all catchments under different lead time. Each catchment is labeled with the best NSE model and its NSE value.

performer, accounting for 7.1 %, 9.9 %, 13.7 %, 13.2 %, and 17.5 % of the top performances at the corresponding lead times. Other Transformer-based variants (excluding PatchTST) generally underperformed compared to both LSTM and Mamba. Among them, Informer yielded better results than RRformer, while Autoformer consistently showed the poorest performance.

Model performance exhibited clear regional variations. In coastal

regions with temperate climates and stable rainfall regimes—such as Oregon, Washington, and North Carolina—all models generally delivered strong predictive accuracy. As the lead time increased, performance gradually declined, with the best-performing models in these regions being primarily Patch TST, LSTM, and Informer. In contrast, models struggled in inland central and southern regions (e.g., Kansas and Texas), where complex terrain and variable climatic conditions led to

lower forecasting accuracy across all lead times; here, Patch TST remained the most frequently optimal model. Similarly, in the Great Lakes region, particularly in Michigan, Patch TST consistently outperformed other models. This may be attributed to the ability of self-attention mechanisms to capture long-range dependencies among meteorological variables, such as wind speed, which play a key role in lake-influenced hydroclimatology.

According to the evaluation scheme by Ritter and Muñoz-Carpena (2013), we conclude that while transformer-based models are capable of achieving state-of-the-art performance in refined runoff and flash flood forecasting, this superiority is highly dependent on architectural design. Specifically, the patch-based Patch TST variant consistently outperformed other models, whereas other transformer architectures generally matched or underperformed relative to the established LSTM benchmark. Thus, the advantage of the transformer paradigm in hydrology is not universal but hinges on appropriate structural inductive biases for temporal hydrological data.

4.2. Approach comparison

To analyze the specific differences between the models, we further compared the NSE distribution of each model at different lead times, as shown in Fig. 4. The bar charts reflect the results of each model across the 516 watersheds, with red lines connecting the medians.

It can be seen that, except for the 1-hour lead time (shown in Fig. 3 (a)), the Patch TST performs best at all lead times, overwhelmingly surpassing other models, while Autoformer shows the poorest performance with a clear skewness. The LSTM and Informer models are relatively close in terms of median, but LSTM exhibits greater variance. Surprisingly, although the Mamba model accounts for a very low proportion among the optimal models, its NSE median is close to that of the

Informer and LSTM models. It is noticed that the NSE gap between the Mamba model and these models is not significant, or that the advantage of these models over Mamba in most watersheds is relatively minor. Dlinear and RRformer also exhibit similar patterns, but the NSE distribution of RRformer is more dispersed than that of Mamba. This indicates that Mamba's overall performance is more stable, while the results of Dlinear and RRformer under extreme conditions are less reliable.

The NSE value reflects the reliability of the model under base flow conditions, but for flood predictions, it is crucial to focus on the peak flow portion. In Fig. 5, the deviation levels of each model in peak flow simulation are further illustrated. A smaller TPE-2 % indicates that the model's prediction and simulation of peak flows are more accurate.

The Patch TST continues to maintain its leading advantage in peak flow simulation, although this advantage gradually diminishes as the lead time increases. Autoformer remains the poorest-performing model. The peak error levels of Informer and RRformer are similar, whereas Dlinear has a lower average error than both. The LSTM model exhibits high uncertainty in peak flow simulation, with significant differences between the lower and upper quantiles. The Mamba model has fewer outliers, further indicating that while its performance may not be the most outstanding, it is relatively stable.

4.3. Interpretable attention mechanism validated by Integrated Gradients

As outlined in Section 2.2.3, the self-attention mechanism in Transformer models computes Query, Key, and Value representations for each element in a sequence, enabling the model to autonomously capture dependencies across different time steps and dynamically weight the relevance of each input. While prior research has established the feasibility of Transformers in hydrological modeling, recent explainable AI (XAI) studies have identified potential problems and limitations

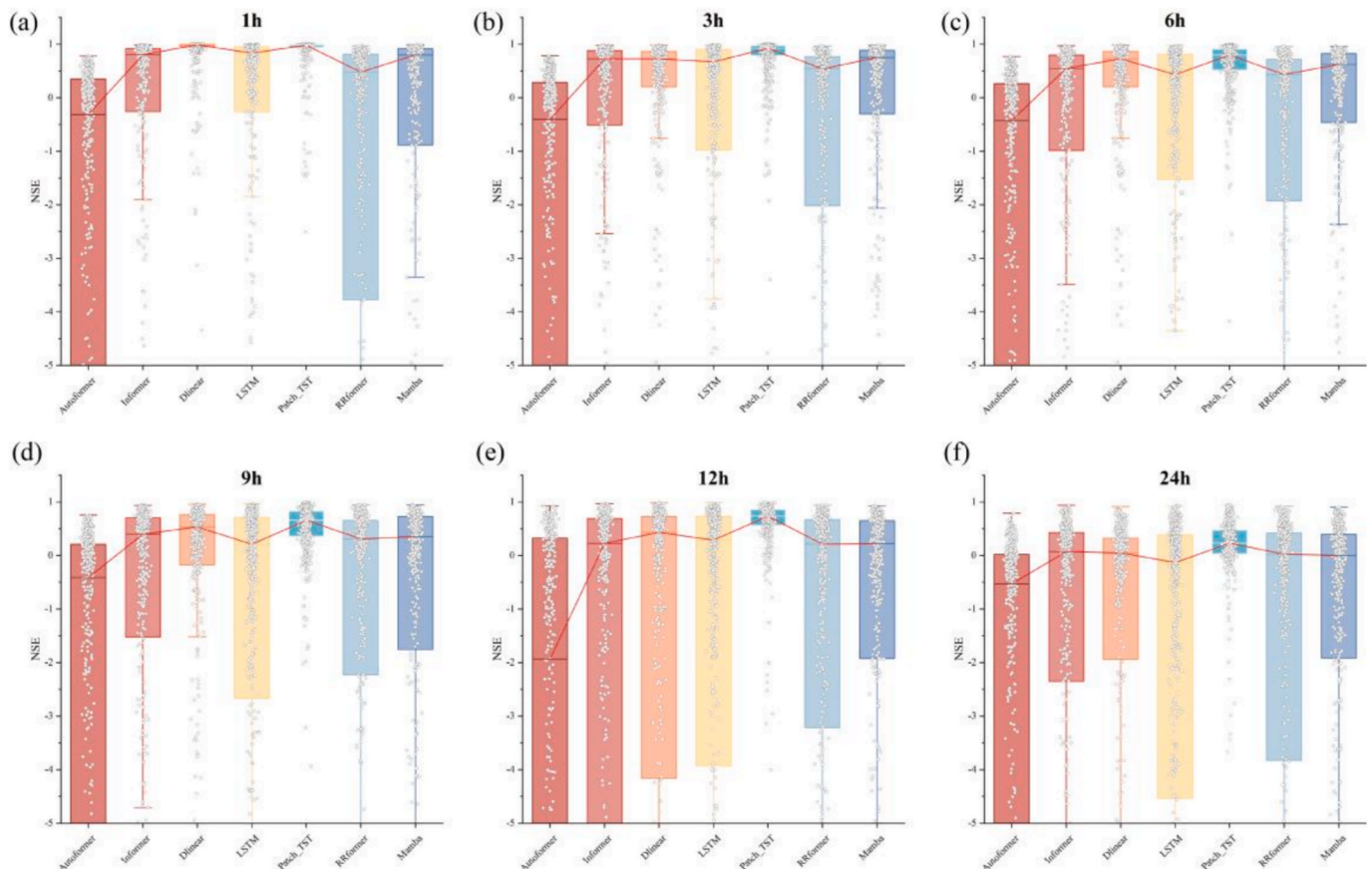


Fig. 4. Box plots of NSE of each model under different lead times.

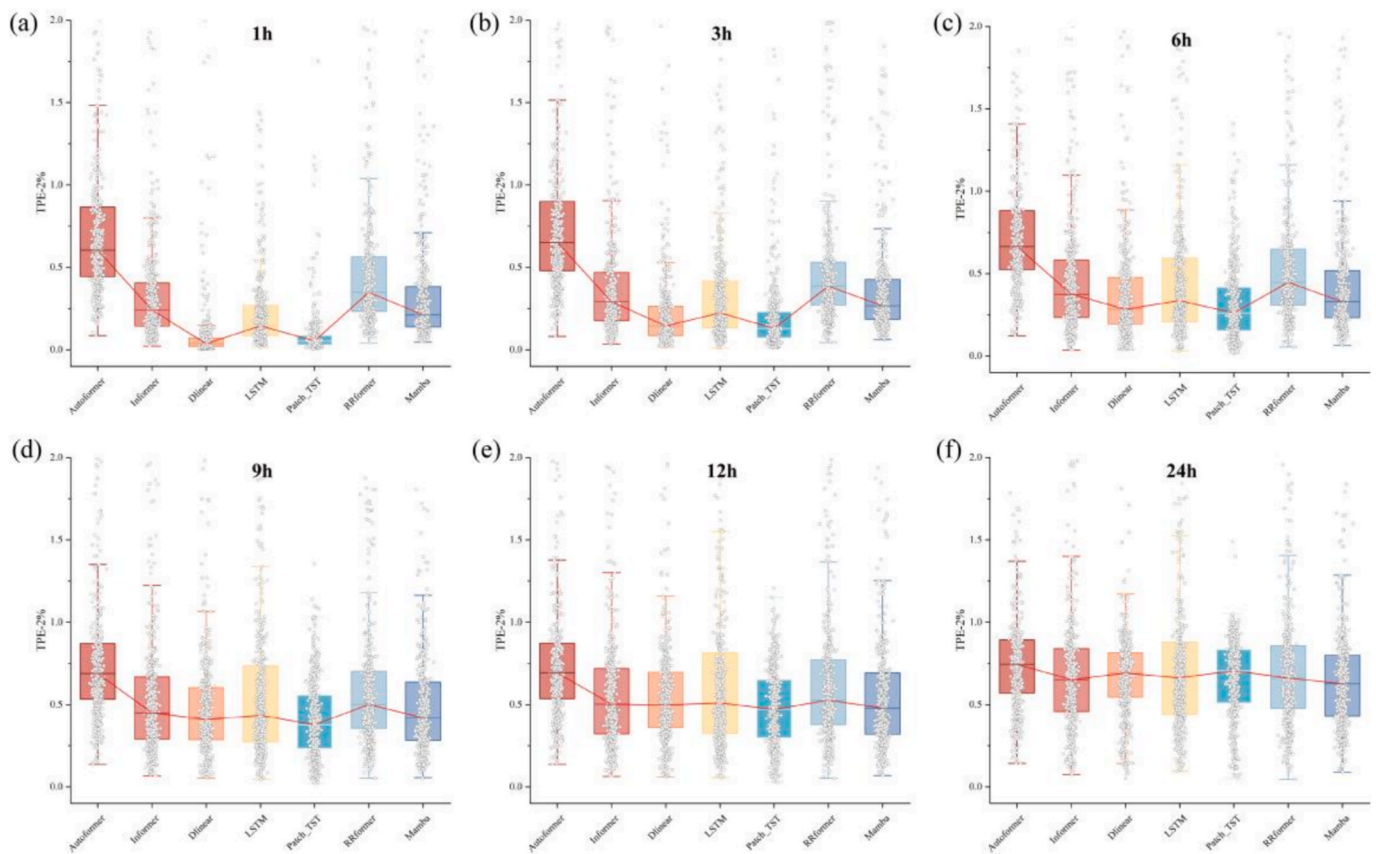


Fig. 5. Box plots of TPE-2 % of each model under different lead times.

inherent to AI explainability (Beven, 2020; Heudorfer et al., 2025). To provide a robust physical interpretation, we move beyond raw attention scores by cross-validating the internal attention allocations with IG—a widely adopted post-hoc attribution method implemented via the PyTorch Captum library (Kokhlikyan et al., 2020). Based on a random sample of 2,000 instances from the test set, we investigate temporal-wise dependencies using the Patch TST and variable-wise attribution using the Informer, providing new insights into how these models prioritize information during hydrological simulation.

4.3.1. Attention in current is greater than earlier and move stably with sequences

The Patch TST model partitions each input sequence into non-overlapping blocks, which are encoded as patches and processed by the encoder. In our experiments, the patch length was set to 8 h, resulting in 12 patches per input sequence covering the preceding 96-hour period. These patches were categorized into recent, medium-term, and long-term segments. Attention scores were computed between every pair of patches, allowing us to quantify the distribution of attention across different temporal ranges. Further computational details are provided in the [supplementary material](#).

In parallel processing architectures, inputs are theoretically processed without inherent sequential bias. However, as illustrated in Fig. 6, the trained model exhibits a distinct macro-temporal preference. When grouped into broad phases, both the raw attention distribution (Fig. 6a) and the IG relative attribution (Fig. 6b) exhibit a consistent overarching temporal decay. Statistical analysis reveals that 36.5 % of the raw attention is allocated to the Current period, followed by 33.5 % to the Recent period, and 30 % to the Early period. The IG method corroborates this macro-trend (44.4 %, 30.2 %, 25.4 %), confirming that recent meteorological conditions exert a stronger overall influence on the predicted runoff than earlier ones, effectively capturing the natural

attenuation of rainfall-runoff processes.

Crucially, at a finer granular level, a notable and theoretically sound divergence emerges between the two metrics. The IG attribution displays a nearly monotonic decay from the most recent patch to the oldest, heavily prioritizing the immediate inputs. In contrast, the raw attention scores highlight a secondary concentration of importance in patches corresponding to the past 24–32 h. This divergence underscores the XAI principle that attention does not strictly equal direct feature attribution. While IG highlights the most direct gradient contributors (typically the immediate autoregressive and precipitation signals), the self-attention weights specifically learn to route delayed information, thereby capturing the lagged response inherent to hydrological systems. We speculate such lags arise from the sequential nature of hydrological processes—including soil infiltration and subsurface routing—which delay the catchment's response.

In summary, the temporal distribution of attention in Patch TST reflects both recency bias and hydrological lag. Moreover, as shown in Fig. 7 using a representative segment from station 1022500, this temporal attention structure remains stable as the input window advances. The dashed black line denotes a rolling historical input window. At the onset of a rainfall event (step 1), attention sharply focuses on the event. As time progresses (steps 2–6), the attention pattern persists but gradually diffuses, demonstrating the model's ability to sustainably track hydrologically meaningful signals over time. We categorize this situation as time-lagged.

4.3.2. Dynamic features attract more attention than static attributes

The benchmark input includes dynamic inputs (meteorological data, historical runoff) and static data (watershed static attributes). Previous studies have utilized various mature methods to conduct sensitivity analyses of different variables on prediction outcomes, such as gradient analysis (Jiang et al., 2022) and SHAP value analysis (Y. Xu et al., 2024).

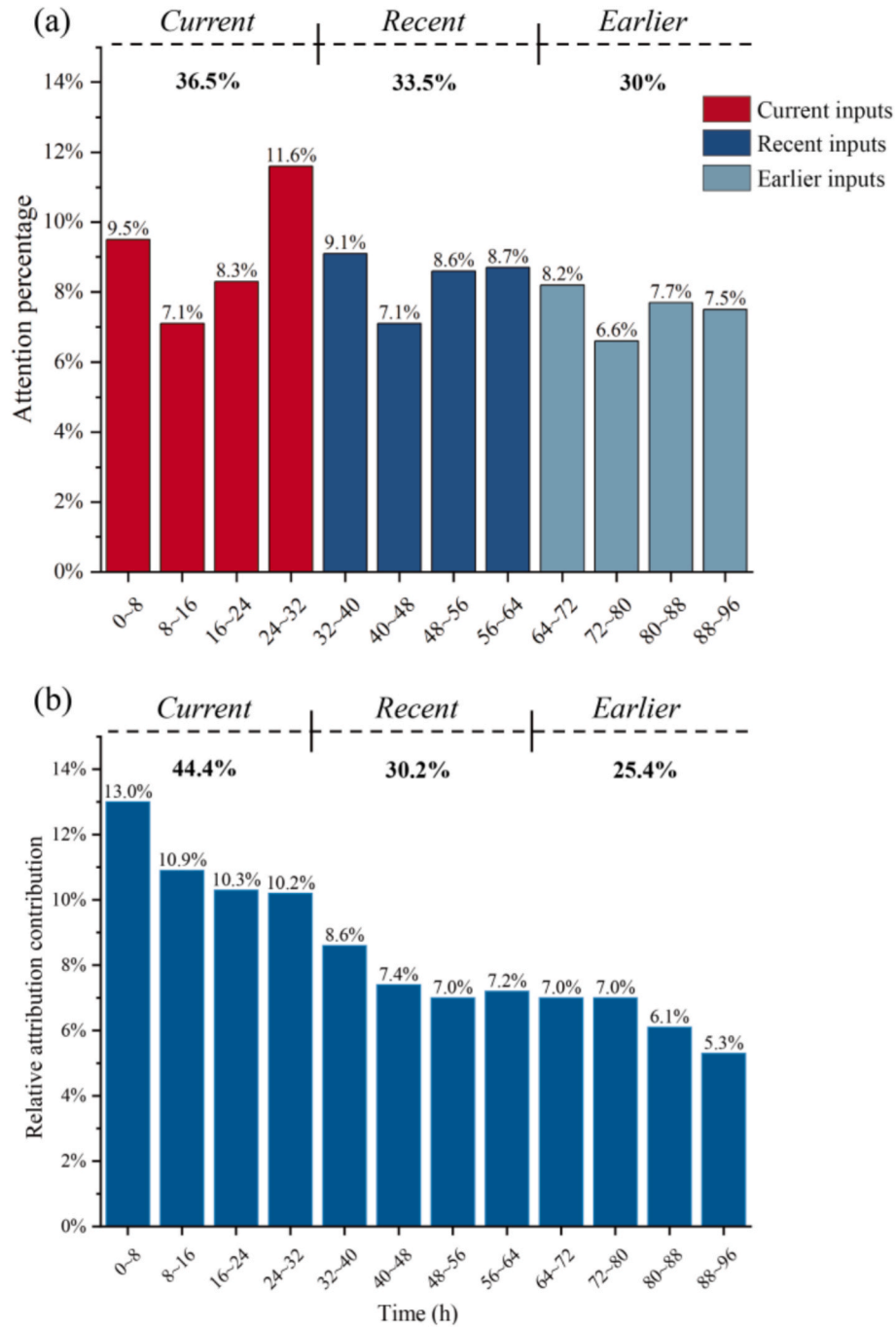


Fig. 6. (a) Proportion of attention scores in the historical input window; (b) Relative attribution contribution by IG methods.

We attempt to analyze input relationships using attention scores. Similar to the statistical methods of attention discussed in Section 4.3.1, we calculated the attention score graphs of the Informer model under the condition of transposed input variables (Y. Liu et al., 2024) and cross-validated them against IG attributions, allowing us to statistically contrast internal attention allocations with actual physical feature importance.

The experimental results indicate that the model allocates higher attention to dynamic variables than to static attributes, although the concatenated input of static attributes can enhance the model's performance in long-term forecasting.

Fig. 8 illustrates the distribution of attention scores and relative IG contributions across different input variables. On a macro level, both metrics exhibit consistent behavioral trends: they overwhelmingly

prioritize time-varying sequences over fixed watershed characteristics. The multi-head attention mechanism assigns an average score of 2.81 % to dynamic variables compared to 2.29 % for static attributes, and both methods highlight precipitation as a highly critical driver.

However, at a finer granularity, profound divergences emerge between attention allocations and direct IG attribution. Notably, IG reveals that historical streamflow provides the most significant physical contribution to the output, acting as a strong autoregressive anchor. In contrast, the attention scores for streamflow are not the highest; instead, attention is heavily directed toward wind speed, rainfall, and humidity. Furthermore, while static attributes attract uniformly low levels of attention in the self-attention module, the IG method clearly identifies varying degrees of physical importance among them. This divergence illustrates that raw attention weights act as internal routing mechanisms

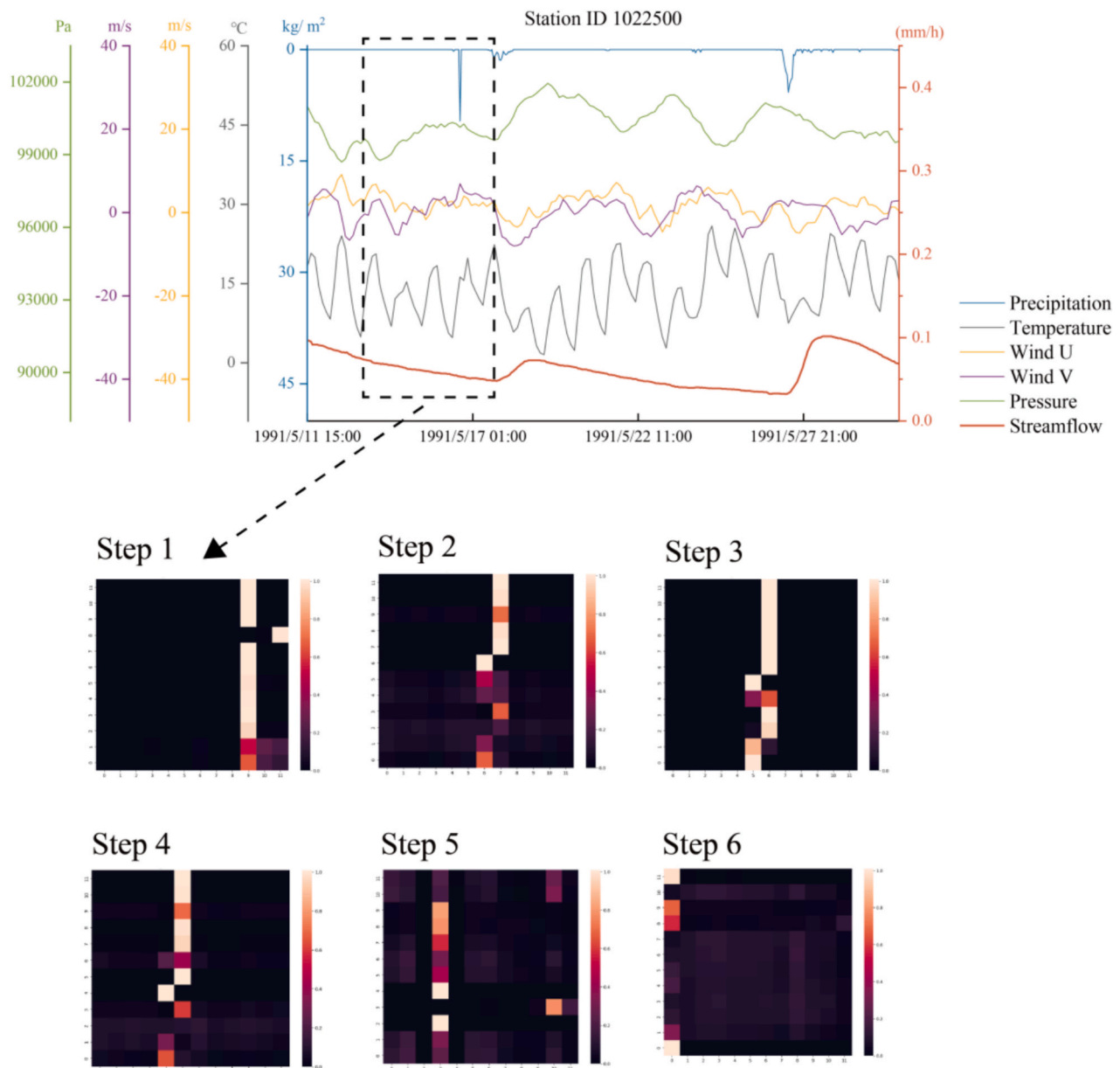


Fig. 7. Attention shift phenomenon with variables at station ID 1022500.

rather than direct measures of feature sensitivity; subsequent nonlinear transformations in the feedforward network further modulate the influence of each variable on the final output, enabling the simulation of complex hydrological processes.

The effective integration of static watershed attributes into deep learning models remains an open research question. Commonly, including in this study, static attributes are concatenated with dynamic variables as a unified input vector. However, our ablation experiments reveal that this simple concatenation approach can in some cases impair model performance. As summarized in Table S3 (supplementary materials), the inclusion of static attributes did not consistently improve short-term forecast accuracy and occasionally degraded it, suggesting potential information redundancy in near-term predictions. In contrast, for longer lead times, most models benefited from static attributes, particularly in simulating peak flow events, where prediction errors were generally reduced.

4.4. Model induction bias under different meteorological and hydrological conditions

Beyond global aggregated metrics, uncovering the specific hydro-climatic conditions under which distinct architectural biases excel provides essential physical insights for hydrological modeling. Building upon the large-sample clustering approach established by Kratzert et al. (2024), we derived six distinct hydro-climatic clusters based on silhouette scores as shown in Fig. 9(a). To facilitate interpretation, we assigned phenomenological labels to these clusters based on the mean characteristics of their constituent variables (detailed in Appendix Table S2): Humid Forested, Lowland Plains, Transitional Regimes, Snowmelt-dominated, Arid and Semi-arid, and Flashy Steep. By analyzing the proportion of catchments where each model achieved the highest NSE score, distinct regional performance patterns emerge.

For ultra-short lead times (e.g., 1 h), the simple layered structure (Dlinear) exhibits overwhelming dominance across most regimes. This finding suggests that for immediate nowcasting, the strong autoregressive signal of historical streamflow effectively overrides complex

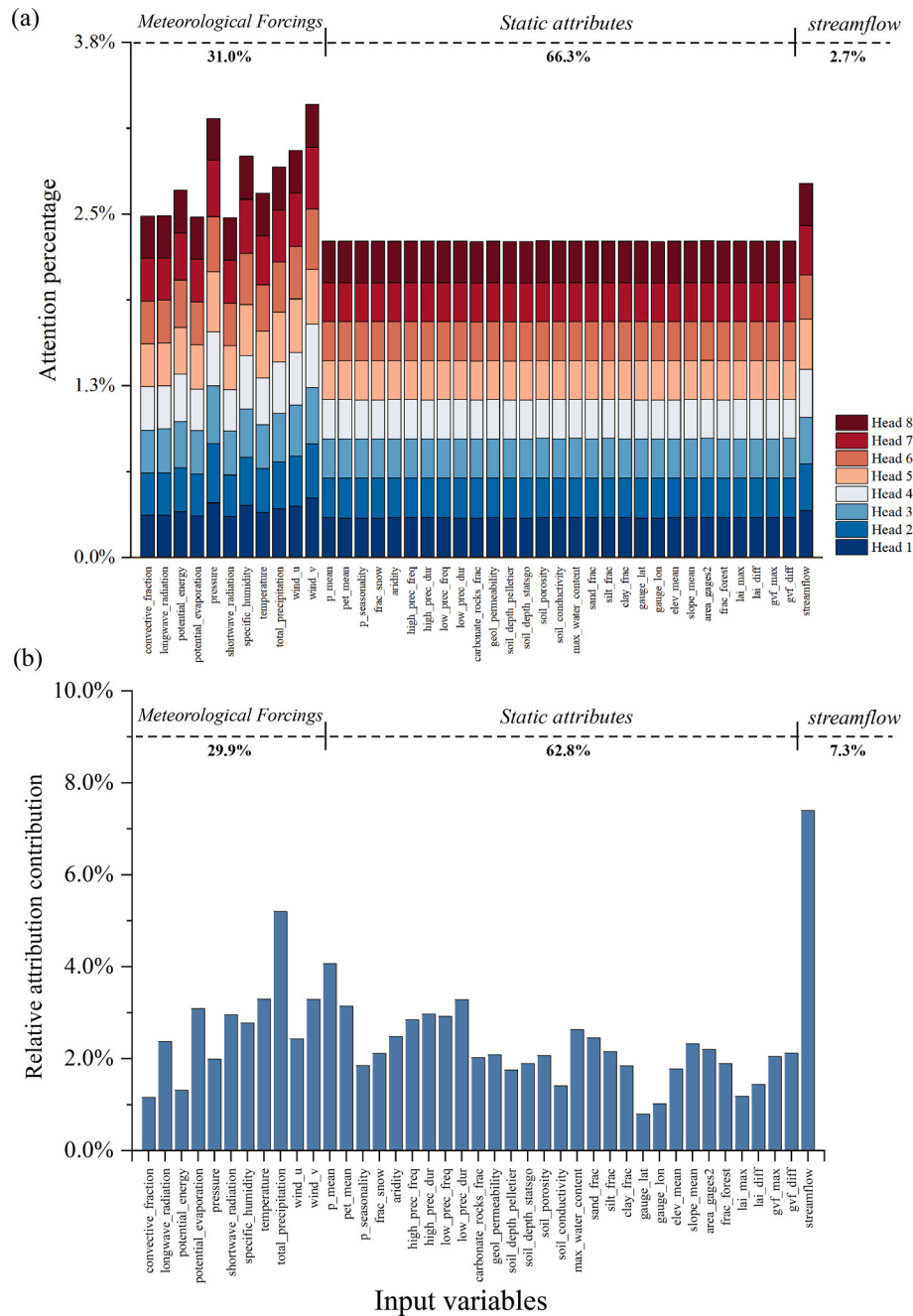


Fig. 8. (a) Proportion of attention scores in different input variables; (b) Relative attribution contribution by IG methods.

hydro-climatic variability, allowing simple linear decomposition to suffice.

However, as the forecast horizon extends, structural inductive biases become the decisive factor. The Patch TST consistently maintains the most robust performance globally. In arid and semi-arid catchments, which are typified by prolonged dry spells punctuated by episodic, intense rainfall, the Patch TST demonstrates a pronounced advantage. Its patch-based processing inherently smooths zero-flow noise while its global receptive field accurately captures distant meteorological triggers. Conversely, in catchments governed by slow, cumulative hydrological processes—such as Humid Forested and Snowmelt-dominated regimes—the LSTM and Mamba models exhibit improved relative performance. This aligns with hydrological intuition: the continuous cell state of LSTM and the selective memory of Mamba inherently mirror the slow mass accumulation and delayed subsurface routing characteristic

of these environments.

Interestingly, in Flashy Steep regimes, characterized by rapid concentration times and non-linear runoff generation, the predictive edge of Patch TST gradually diminishes at longer lead times. This trend indicates an inherent predictability limit in such flashy basins, where all evaluated architectures struggle to maintain high accuracy without real-time data assimilation. Furthermore, it is noteworthy that other attention-based models (e.g., Informer, Autoformer), while initially lagging behind Dlinear and LSTM, begin to reveal their capacity for long-term sequence extrapolation as the forecasting horizon extends, eventually surpassing simpler baselines in extended routing scenarios.

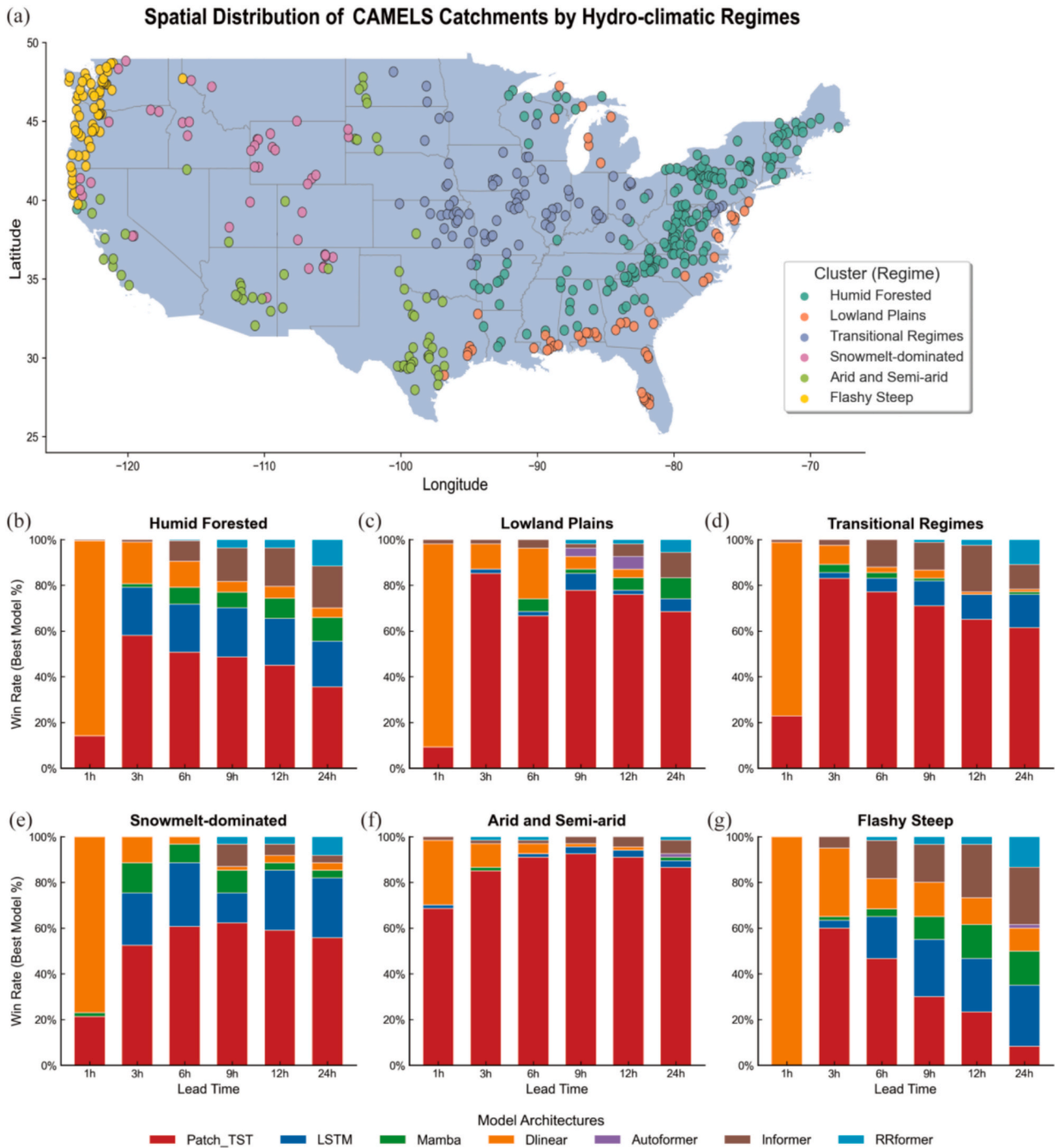


Fig. 9. Spatial distribution of CAMELS catchments across hydro-climatic regimes and model performance dominance at different lead times.

4.5. Disentangling discharge-assisted forecasting from Rainfall-Runoff forecasting

As emphasized previously, our primary focus is on discharge-assisted forecasting, which is highly relevant for operational short-term flood early warning. However, including historical streamflow inherently conflates the routing of existing channel flow with the physical mapping of meteorological forcings to runoff. To rigorously isolate the models' fundamental rainfall-runoff learning capabilities, we conducted an additional comparative benchmark excluding historical streamflow (relying strictly on meteorological forcings and static attributes).

Fig. 10 illustrates the empirical Cumulative Distribution Function (CDF) of the NSE scores across various lead times for both scenarios.

Solid lines represent the discharge-assisted forecasting scenario (with historical streamflow), while dashed lines denote the pure rainfall-runoff forecasting scenario (without historical streamflow).

Expectedly, the exclusion of historical streamflow leads to a drastic degradation in performance across all architectures. Without the strong autoregressive signal to anchor initial conditions, the models are forced to reconstruct the complex antecedent moisture and subsurface states purely from sequence history. This severe performance drop fundamentally limits the operational viability of pure rainfall-runoff deep learning models at the hourly scale, thereby strongly validating the rationality of incorporating historical streamflow in hourly flood prediction.

Interestingly, beneath this general degradation, the architectural

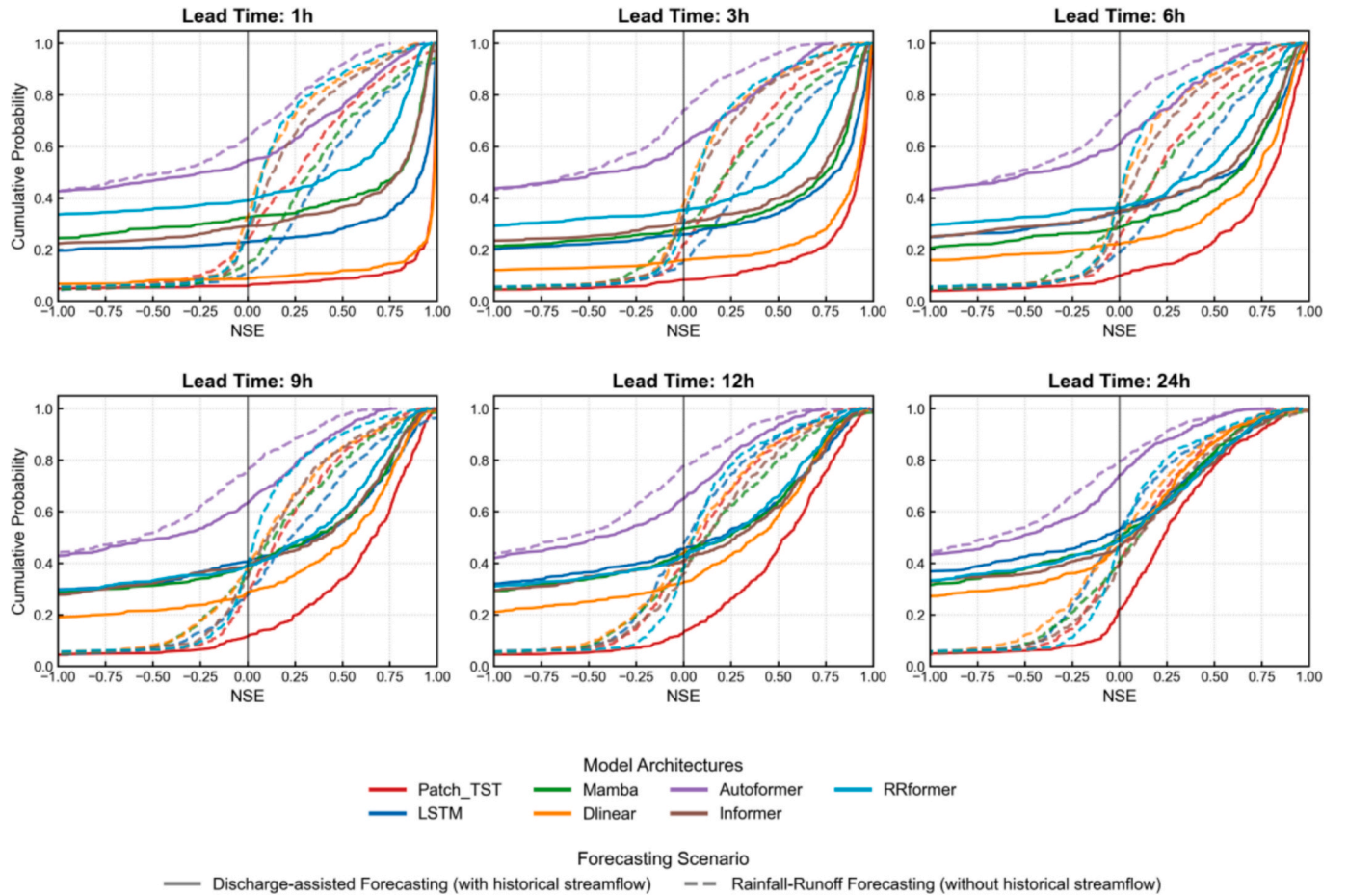


Fig. 10. Cumulative Distribution Function (CDF) of NSE scores under discharge-assisted and pure rainfall-runoff forecasting scenarios.

ranking undergoes a notable shift. While Patch TST overwhelmingly dominates the discharge-assisted task—leveraging its superior sequence extrapolation capabilities when an initial flow trajectory is provided—its absolute superiority diminishes in the pure rainfall-runoff setting. Under these strict forcing-to-discharge conditions, recurrent and state-space architectures (LSTM and Mamba) exhibit remarkable resilience. This divergence highlights a critical theoretical distinction: patch-based attention provides an optimal inductive bias for routing and sequence extrapolation, whereas recurrent cell states and selective memories are intrinsically more robust for deducing the underlying hydrological process from meteorological inputs alone.

5. Conclusions

This study investigates the temporal inductive biases of diverse deep learning paradigms—including recurrent, attention-based, state-space, and layered architectures—for hourly runoff and flood forecasting. This framework explicitly distinguishes discharge-assisted forecasting from pure rainfall-runoff forecasting and evaluates the applicability of the Mamba (state-space) architecture in high-frequency hydrological modeling.

The LSTM model has long been the primary approach for hydrological time series modeling, largely because its memory mechanism aligns naturally with watershed storage dynamics. However, our experiments demonstrate that model suitability depends heavily on the forecasting task and the underlying hydro-climatic condition. Simple layered structures (e.g., Dlinear) proved highly effective for ultra-short-term forecasting, serving as a robust baseline. As forecast horizons extend, structurally adapted Transformers—particularly those

employing patch-based input representations (e.g., Patch TST)—consistently outperform other models, especially in arid, semi-arid, and flashy steep catchments. This suggests that patching provides a critical inductive bias for sequence extrapolation when historical flow is available. Conversely, in pure rainfall-runoff settings without autoregressive inputs, or in humid and snowmelt-dominated regimes governed by slow subsurface accumulation, recurrent (LSTM) and state-space (Mamba) models are intrinsically more reliable for learning hydrological process from meteorological forcings. Regarding the state-space paradigm, our multi-dimensional assessment indicates that while the Mamba architecture provides stable cross-regime adaptability (as shown in Fig. S2), its overall predictive metrics remain close to, but slightly weaker than, those of the standard LSTM. Nevertheless, it serves as a competitive non-recurrent alternative. Furthermore, we found that concatenating static catchment attributes directly into model inputs increases complexity without consistently improving near-term forecasts, though it proved beneficial for reducing peak flow errors in longer horizons.

Finally, through a multi-perspective interpretability analysis combining self-attention weights with IG, we observed that Transformers learn feature hierarchies consistent with physical hydrological principles. While both the IG method and self-attention weights exhibit a consistent overall trend of temporal decay, the attention mechanism demonstrates a unique capacity to internalize the time-lagged responses inherent in catchment routing. Both metrics agree in prioritizing dynamic meteorological variables over static properties. Future work should focus on refining these architectural inductive biases, and we encourage further research into attention-based architectures for flood modeling, as their ability to directly model interactions across all time steps offers strong representational capacity, scalability to larger

datasets, and promising pathways for hybridization with process-based models (Zhao et al., 2024). Such integration could ultimately enhance forecast accuracy and support more reliable watershed management and decision-making.

CRedit authorship contribution statement

Binlan Zhang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chaojun Ouyang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Qingsong Xu:** Writing – review & editing, Software, Methodology, Conceptualization. **Yanlin Liu:** Writing – review & editing, Methodology, Data curation. **Zili Wang:** Writing – review & editing, Methodology, Data curation. **Quan Tang:** Writing – review & editing, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research is funded by the Sichuan Science and Technology Program (2025YFNH0009), Strategic Priority Research Program of the Chinese Academy of Sciences (Grant No. XDB1390300), West Light Foundation of The Chinese Academy of Sciences (xbzg-zdsys-202401), Chengdu-CAS Science and Technology Cooperation Special Project.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2026.135727>.

Data availability

The data that has been used is confidential.

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